Please note that this preprint has not yet been published in a peer-reviewed journal.

Do you *really* know? Incidental exposure to pro- and anti-consensus science information on social media, knowledge overconfidence, and science skepticism

Yannic Meier

Social Psychology: Media & Communication, University of Duisburg-Essen, Duisburg, Germany

Author Note

Yannic Meier https://orcid.org/0000-0002-5726-3229

I have no conflicts of interest to disclose.

This manuscript features online supplementary material including the items of the study, the dataset, study materials, and additional analyses: https://osf.io/nezwx/?view_only=9b5f2ae5ef9f4b21bdfbdbc7f87b332c

Correspondence concerning this article should be addressed to Yannic Meier, Social Psychology: Media and Communication, University of Duisburg-Essen, 47057 Duisburg, Germany, Forsthausweg 2. E-Mail: yannic.meier@uni-due.de

Abstract

On social media, users are normally not actively seeking content but they are rather incidentally exposed to it. This study examines whether persons with a skeptical disposition towards science are exposed to more information that contradicts scientific consensuses than to information that reflects it. Consequently, users might have the subjective feeling of being well informed about science although they are not. The results of the present preregistered online study ($N = 1279$) indicate that persons who are skeptical towards science are exposed more frequently to both pro- and anti-consensus science information – possibly because they are algorithmically classified as interested in science. Science skepticism and exposure to anti-consensus information are positively associated with knowledge overconfidence whereas pro-consensus information exposure positively relates to objective knowledge. The study suggests that science skeptics are not exclusively exposed to attitude-consistent information which might partially counteract misinformedness. Moreover, even incidental contact with pro-consensus science information can seemingly contribute to objective science knowledge.

Keywords: Incidental Exposure, Science Skepticism, Knowledge Overconfidence, Science Information, Social Media

Do you *really* know? Incidental exposure to pro- and anti-consensus science information on social media, knowledge overconfidence, and science skepticism

On social media platforms like Facebook, YouTube, TikTok, or X (formerly Twitter), users can find an increasing amount of information about scientific topics which are based on scientific consensuses (Pavelle & Wilkinson, 2020). Examples comprise texts, pictures, or videos that explain the anthropogenic global warming, the effectiveness of vaccinations, or the safety of genetically modified food. Most exposures to such science contents on social media sites are, however, not planned and intentional but incidental which means that users come across science information when they had different objectives for using social media (Anderson et al., 2021; Huber et al., 2019; Taddicken & Krämer, 2021). Although there is vast evidence that incidental exposure to political content can positively affect people's political knowledge and content engagement (Nanz & Matthes, 2022), there is only scarce evidence whether these effects translate to science content. Moreover, it is still debated whether people's actual knowledge increases (i.e., objective knowledge) or only their feeling of being well informed (i.e., subjective knowledge) as a consequence of incidental exposure (Schäfer, 2020; Yamamoto & Yang, 2022). Another issue is that not only information which supports scientific consensuses is spread across social media platforms, but also misinformation that unjustifiably contradicts expert consensuses (see Allgaier, 2019) and might, therefore, erode trust in science and even fuel science skepticism and denial (Southwell et al., 2022; Treen et al., 2020). Thus, both exposure to pro-consensus science content and – to an even greater extent – exposure to anti-consensus content could be associated with an overestimation of what one actually knows about different science topics. This is problematic in several ways, for instance, because people who overestimate their objective science knowledge have more negative attitudes towards science, deny scientific consensuses, and tend to spread misinformation (Ho et al., 2019; Light et al., 2022; Williams Kirkpatrick, 2021).

There are several possibilities why social media users are incidentally exposed to

certain contents. For example, they might have subscribed to channels, social contacts might share content, or content is algorithmically recommended (Thorson & Wells, 2016). All these factors might contribute to users seeing more content which matches rather than contradicts their attitudes. This raises the question of whether persons with negative attitudes towards science are exposed more frequently to misleading than accurate science content – as a possible consequence of their network and past interactions with such contents (Rousseau, 2023). The present study has the objective of examining whether social media users who are skeptical about science are incidentally exposed to more anti- than pro-consensus science content and whether incidental exposure to pro- or anti-consensus content and science skepticism are associated with overestimating ones' objective science knowledge. The study helps to understand whether dispositions towards science are linked to incidental exposure to science content on social media platforms and to false perceptions of one's knowledge.

Literature Review

Science (Mis)information on Social Media

Social media users are incidentally exposed to various kinds of information, for instance, political information (Nanz & Matthes, 2022), current news (Oeldorf-Hirsch, 2018), health information (Tian & Robinson, 2009), or science information (Anderson et al., 2021). Such incidental exposures occur when users encounter certain contents unintentionally and pursue a different goal such as seeking entertainment or sports (Greene et al., 2021; Tewksbury et al., 2001). These unintentional contacts can happen when subscribed channels disseminate new contents, when social contacts share contents, or when contents are algorithmically recommended (Thorson & Wells, 2016). On social media, not only accurate science information which reflects expert consensuses is spread but also vast amounts of misinformation which contradict these consensuses (Allgaier, 2019). Vraga and Bode (2020) define misinformation by the amount of expert consensus and the amount of available evidence. Information which contradicts a given consensus

based on a large amount of evidence is always referred to as misinformation because it unjustifiably contradicts both experts and evidence. In this study, I will use the terms anti-consensus information and science misinformation interchangeably to describe information which contradicts the scientific consensus in different fields.

While incidental exposure to news and political information has received significant scholarly attention, incidental exposure to scientific content received comparatively little attention. However, political news and science content differ in that political content oftentimes comprises current events and questions without a definite right-or-wrong answer (e.g., whether or not to legalize marijuana). Science information, on the contrary, presents recent empirical evidence or expert consensuses and, therefore, the best available evidence at a time (Vraga & Bode, 2020). Also, science information often has implications for people's decision making, for example, whether to get vaccinated, whether genetically modified foods are safe to eat, or how to live in a climate- and environment-friendly way. Hence, I argue that science information has a particular status for people, and they might pay different attention when incidentally exposed to it compared to news and political content which might be evident in generally high levels of trust in science around the globe (Cologna et al., 2024).

Objective and Subjective Knowledge

When users are motivated to engage in cognitively effortful processing of content they are incidentally exposed to, they might store these contents in their memory – irrespective of whether this content delivers accurate or false information (Greene et al., 2021; Matthes et al., 2020). However, when people only engage in shallow content processing like scanning a headline, objective knowledge acquisition is unlikely (Chen et al., 2022; Matthes et al., 2020), but someone might have the impression of having learnt something new (Plohl & Musil, 2018). Moreover, persons who encounter misinformation might also feel well-informed. Thus, incidental exposure has the potential of evoking the feeling of being well informed – even when people are factually not or believe false claims

(Anspach et al., 2019; Lee et al., 2020). These two facets of knowledge have been termed objective and subjective knowledge (Brucks, 1985). The objective component refers to the actual information that is stored in people's memory. The subjective component describes people's perception and assessment of their levels of knowledge, that is how much they believe to know. While objective knowledge informs about the memory contents of persons and can, thus, be measured objectively, subjective knowledge is a self-assessment of one's knowledge and can, therefore, indicate a variety of things ranging from people's self-confidence, heuristics and cognitive biases (Park & Lessig, 1981), to levels of misinformedness (Partheymüller et al., 2022).

Investigating both people's objective and subjective knowledge about various science topics is crucial as the two knowledge facets appear to be related to science attitudes and the acceptance of science in different ways. Subjective science knowledge, for instance, has been found being associated with a reduced support for science and technology funding (Ho et al., 2019) and with an increased likelihood to share articles containing fake science claims on social media (Williams Kirkpatrick, 2021). Moreover, Light et al. (2022) demonstrated that persons with the widest gap between objective and subjective scientific knowledge concerning various scientific consensuses are also those who most strongly reject the scientific consensus in these areas. In contrast, objective science knowledge is associated with positive attitudes towards science and scientists and with a greater acceptance of scientific consensuses (Ho et al., 2019; Light et al., 2022; Takahashi & Tandoc, 2016). Consequently, it is essential to study how exposure to science information on social media is related to both user's objective and subjective science knowledge.

Incidental Exposure to Science Content

More frequent incidental exposure to pro-consensus science content should positively affect users' objective knowledge (Greene et al., 2021), however, research in this area is still scarce. Anderson et al. (2021) found no support for their hypothesis that Facebook users who are more frequently incidentally exposed to information about genetic modification

have higher objective knowledge about genetic modification. In an experimental study, however, Anspach et al. (2019) were able to show that confronting participants with a social media posting about genetically modified foods can increase their knowledge, but not as much as an in-depth article. This indicates that social media science postings have the potential to contribute to users' objective science knowledge, but only to a limited extent as they are often less detailed than other online contents. However, a meta-analysis about incidental exposure to political information has revealed that effect-sizes with knowledge in experiments tend to be larger compared to surveys (Nanz & Matthes, 2022). Hence, more research is required to evaluate whether incidental exposure to pro-consensus science content can contribute to users' objective science knowledge or whether users' subjective knowledge increases in the first place. For example, more frequent incidental exposure to news or political contents has been found to be associated with increased subjective knowledge (Yamamoto & Yang, 2022). Moreover, Schäfer (2020) demonstrated that being exposed to a high number of social media news posts is associated only with increased subjective knowledge while only those who fully read a news article had a gain in objective knowledge. This could indicate that incidentally exposed content is primarily scanned superficially rather than effortfully processed with the latter being a precondition for gaining objective knowledge (Chen et al., 2022; Matthes et al., 2020; Nanz & Matthes, 2022). Another experimental study showed that reading a text about nanotechnology that does not contain any helpful information about nanotechnology but only the term can cause an illusion of knowledge (Plohl & Musil, 2018). Thus, social media users might get the impression of being well informed about the topics they are regularly exposed to which is not necessarily the case when they only read headlines or posts with reduced information, for example.

While empirical evidence about whether incidental exposure to pro-consensus science information can increase users' science knowledge is scarce, there appears to be even less evidence regarding anti-consensus information. When people are frequently

exposed to anti-consensus information, their objective knowledge might decrease if they start to believe false claims because, hypothetically, people can learn from incidental exposure to false and misleading information (Greene et al., 2021). Lee et al. (2020) found that persons who were more frequently exposed to Covid-19 misinformation were more likely to believe false claims about Covid-19. Moreover, Nyhan et al. (2022) found that although exposure to pro-consensus climate science information can increase people's climate science knowledge, these effects can be rapidly reversed by exposure to information which challenges the scientific consensus. Together, these studies indicate that anti-consensus information has the potential to affect people's objective knowledge in a negative way. Considering the paucity of empirical evidence regarding incidental exposure to pro- and anti-consensus science information, I assume that exposure to pro-consensus content will be positively associated with both objective and subjective knowledge whereas exposure to anti-consensus content will be negatively related to objective but positively to subjective knowledge. Accordingly, the following two hypotheses are proposed:

Hypothesis 1 (H1): More frequent exposure to pro-consensus information on social media is a) positively associated with subjective science knowledge and b) positively associated with objective science knowledge.

Hypothesis 2 (H2): More frequent exposure to anti-consensus information on social media is a) positively associated with subjective science knowledge and b) negatively associated with objective science knowledge.

Knowledge Overconfidence

Sometimes, people overestimate how much factual knowledge they possess. Especially persons with low abilities or little objective knowledge seemingly tend to overestimate what they are capable of and how much they know (Kruger & Dunning, 1999; Plohl & Musil, 2018). People's overestimation of knowledge levels can be explained by an inability to recognize one's own low levels of competencies. Currently, it is debated whether incidental content exposure contributes to such illusions of knowledge (Schäfer,

2020; Yamamoto & Yang, 2022). Being exposed to a high number of social media news posts has been found to be associated only with increased subjective knowledge while only those who fully read a news article had a gain in objective knowledge (Schäfer, 2020). Regarding science information, Anspach et al. (2019) found that especially social media postings which only contain a reduced amount of information compared to, for example, full articles have the potential to contribute to overestimating one's objective knowledge. The authors argue that knowledge overconfidence is a form of misinformedness, that is people's level of confidence about holding false beliefs (in contrast to being uninformed meaning that people are aware of their deficient knowledge). As social media users might have objective knowledge gains from being incidentally exposed to accurate pro-consensus science information, especially anti-consensus information exposure should be associated with users' knowledge overconfidence (see Nyhan et al., 2022). Consequently, the following hypothesis and research question are derived:

Hypothesis 3 (H3): More frequent exposure to anti-consensus information on social media is positively associated with knowledge overconfidence.

Research Question 1 (RQ1): Will frequent exposure to pro-consensus information on social media be associated with knowledge overconfidence?

General Science Skepticism

Incidental exposure is unintentional but not coincidental (Thorson, 2020). Users can partly customize their feeds by subscribing to the channels they like which increases the likelihood to be exposed to content from that channel in the future and social contacts may share contents from channels someone does not follow themselves or create novel content (Thorson & Wells, 2016). Because social media companies track users' (intentional) behaviors, algorithmic inferences about user interests are used to display content in the future (DeVito, 2017; Sabir et al., 2022). To illustrate, Thorson et al. (2021) found that Facebook users who had been algorithmically classified as interested in news and politics were also more likely to be exposed to these contents. Such classifications can be the result

of behaviors such as liking, commenting, reading, or simply clicking on links all of which have been found to be associated with increased incidental exposure (Ahmadi & Wohn, 2018; Barnidge, 2023; Oeldorf-Hirsch, 2018; Rousseau, 2023). Longitudinal works have further corroborated these claims (Barnidge, 2023; Rousseau, 2023). For example, Rousseau (2023) found that more engagement (e.g., liking, commenting, or sharing) with climate-science content leads to higher incidental exposure to climate-science content after a couple of months. Altogether, incidentally exposed content is likely to reflect users' interests and dispositions due to customization, social contacts, and algorithmic recommendations.

Because small parts of western societies hold negative attitudes towards science and scientists and refute scientific consensuses (Lewandowsky et al., 2013), it is an important question to study whether these individuals see more science misinformation on social media which may potentially reinforce their skepticism and denial (Southwell et al., 2022; Treen et al., 2020). In the present study, we turn to general science skepticism because this distrusting and doubting disposition towards science and scientist has been found being positively associated with the rejection of various scientific consensuses such as global warming, the advantages of vaccinations, or the severity of Covid-19 (Meier & Krämer, 2022). Moreover, science skeptics were found having substantially lower trust in science information presented in the media (Meier & Krämer, 2022). Such reduced trust may be associated with different kinds of social media behaviors like avoiding pro-consensus content, leaving negative comments under pro-consensus posts, or sharing opposing beliefs (see Walter et al., 2018). Moreover, users might intentionally subscribe to channels that share anti-consensus information which may subsequently lead to more incidental exposure to similar content. Based on these considerations, science skeptics might also be algorithmically classified as interested in anti-consensus information (Thorson et al., 2021) and, therefore, might be exposed to such contents more frequently (Ahmadi & Wohn, 2018; Barnidge, 2023; Oeldorf-Hirsch, 2018). So far, however, there seems to be only very little

research pursuing the questions which interindividual factors are associated with incidental exposure to pro- and anti-consensus science information. Finally, because they are more likely to refute scientific consensuses, persons who are skeptical about science might possess less accurate science knowledge (see Meier & Krämer, 2022) and overestimate their levels of objective knowledge (Light et al., 2022). Consequently, the last two hypotheses are proposed:

Hypothesis 4 (H4): General science skepticism is a) negatively associated with the frequency of being exposed to pro-consensus information on social media and b) positively associated with the frequency of being exposed to anti-consensus information on social media.

Hypothesis 5 (H5): General science skepticism is positively associated with knowledge overconfidence.

Method

Open Science

The present study was preregistered on the platform of the Open Science Framework: https://osf.io/3daxu?view_only=5f9140807bcb4022a3e5856821070e80. Moreover, the dataset, R-code, and survey items were uploaded to the OSF: https://osf.io/nezwx/?view_only=9b5f2ae5ef9f4b21bdfbdbc7f87b332c.

Sample

This study is part of a larger project for which data of 1610 German internet users were collected. An ISO 20252:2019 certified online panel provider was contracted for recruiting and incentivizing participants. Two hundred and thirty-six persons who stated to not use social media were excluded from the analyses. Furthermore, 95 participants were excluded who completed the survey in less than 6 minutes which was considered as unrealistically fast. Thus, the final sample comprised 1279 social media users (650 women, 626 men, 3 other) aged 18 – 74 ($M = 44.64$, $SD = 15.14$). Most participants had a medium German school degree ($n = 383$), followed by persons with a university degree ($n = 339$),

and individuals with a university entrance qualification (n = 309). The most used social media sites were YouTube (n = 1001), Facebook (n = 864), Instagram (n = 764), and TikTok (n = 307).

Measures and Procedure

The scales are presented in the same order as they were presented in the survey. All items were randomized. The items covering subjective knowledge, objective knowledge, and the frequency of contact to pro- and anti-consensus information concern the following seven topics: genetically modified food, vaccinations, global warming, homeopathy, the big bang theory, future pandemics, and the theory of evolution. These topics have been selected in accordance with previous studies (Light et al., 2022; Meier & Krämer, 2022) since for every of these topics, there is extensive evidence pointing to an expert consensus[1].

Participants' *subjective knowledge* about the seven science topics was assessed by means of seven items on a seven-point Likert-scale ranging from 1 = *very little knowledge* to 7 = *very much knowledge*. The items were based on those of Light et al. (2022) started with "How would you rate your level of knowledge about…" followed by the respective topic, for instance, "…vaccinations".

Objective knowledge was assessed by means of 21 items, three per topic. Eight of these items were taken from previous studies (Betsch & Wicker, 2012; Light et al., 2022; Tobler et al., 2012) and 13 were developed for the purpose of this study. There were ten correct statements (e.g., "Humans share most of their genes with chimpanzees") and eleven incorrect statements (e.g., "Vaccinations increase the occurrence of allergies"). The response options were "true", "wrong", and "I don't know" with correct answers being coded as 1, and the other two options as 0. Hence, every participant had a score ranging from 0 (no correct answer) to 3 (only correct answers) for every of the seven science topics.

Incidental exposure to pro- and anti-consensus information was assessed with 14

[1] We have uploaded a document with references pointing to expert consensuses in all seven science domains to the OSF: https://osf.io/nezwx/?view_only=9b5f2ae5ef9f4b21bdfbdbc7f87b332c.

original items starting with the sentence "Referring to the last three months, how often were you shown posts on social networks - without actively searching for them - that ..." followed by the respective topic and direction, for instance, "describe vaccinations as [safe/unsafe] and [beneficial/disadvantageous]". The possible answers ranged from 0 = *never* to 7 = *several times a day* (please see the online supplement for the full set of response-options).

General science skepticism was measured by means of 18 items developed by Meier and Krämer (2022) on a seven-point Likert-scale ranging from 1 = *I do not agree at all* to 7 = *I fully agree*. An example item is "Scientists act with questionable intentions". The reliability of the scale was assessed by means of a confirmatory factor analysis that revealed two higher order factors in line with those reported by Meier and Krämer (2022). The CFA revealed a good model fit: $\chi^2(134) = 409.69$, $p < .001$, $\chi^2/df = 3.06$, CFI = .98, TLI = .98, RMSEA = .040 (90% CI: .037, .044), SRMR = .03, and two reliable and internally consistent factors (factor 1: Cronbach's $\alpha = .96$, McDonald's $\omega = .96$, AVE = .71; factor 2: Cronbach's $\alpha = .92$, McDonald's $\omega = .92$, AVE = .60).

Knowledge overconfidence was calculated from participants' subjective and objective knowledge scores using the same procedure as (Light et al., 2022). First, both variables were z-scored to make them comparable. Afterwards, participants' objective knowledge was subtracted from their subjective knowledge. Accordingly, positive values indicate an overconfidence (i.e., participants' subjective knowledge exceeds their objective knowledge), values around zero indicate a realistic assessment of one's knowledge, and negative values indicate an underestimation of one's knowledge (i.e., participants' subjective knowledge is lower than their objective knowledge).

Statistical Power

A smallest effect size of interest (SESOI; Lakens, 2022) was used to a) estimate an appropriate sample size and to b) have a second inference criterion besides the p-value. The SESOI was set to a small, standardized effect such as a Pearson's r of .10 or a

standardized regression coefficient (β) = .10. Effects lower than that will not be interpreted as meaningful. The structural equation model was simulated using an alpha-level of 5%, the SESOI, and the current sample size (N = 1279). The likelihood to find effects as small as the SESOI is approximately 95%.

Results

All statistical analyses were carried out with R-Studio and R version 4.3.2. Zero-order correlations of all variables can be seen in Figure 1 and descriptive statistics are reported in the following. Participants had a rather medium to low science skepticism (M = 3.5, SD = 1.3), answered an average of eleven of the 21 science statements correctly (M = 11.14, SD = 4.47), and had a medium subjective knowledge (M = 4.07, SD = 1.2). Moreover, they reported being exposed to both pro-consensus science information (M = 2.22, SD = 1.39) and anti-consensus information (M = 2.02, SD = 1.38) about once a month for each of the seven topics. Finally, on average, respondents were neither over- nor underconfident about their scientific knowledge (M = 0.00, SD = 0.8).

To test the first two hypotheses, correlation analyses were calculated. Due to space constraints, correlation coefficients are not reported in the text. Please see Figures 2 and 3 for a complete overview. Hypothesis H1a assumed that more frequent exposure to pro-consensus information would be positively related to subjective knowledge in the seven science domains. There are positive correlations between exposure to the seven science topics and subjective perceptions of one's knowledge in the respective areas with small to medium-sized relations. Consequently, *H1a* was supported by the data. *H1b* suggested that exposure to pro-consensus information would positively relate to objective knowledge in the seven science domains. This hypothesis was only partly supported as only four significant (of the seven) relationships were above the SESOI (i.e., $r > .10$) while two significant relations were below the SESOI, and another non-significant relationship was found. In *H2*, I hypothesized that more frequent exposure to anti-science information would a) positively relate to subjective science knowledge and b) negatively to objective

science knowledge. H2a was supported by the data since there were positive significant correlations above the SESOI in all seven domains, again, revealing small to medium relationships. In contrast, the data did not support H2b since six of the correlations were not significant and the only significant correlation was in the opposite direction. In sum, these findings indicate that incidental exposure to pro-consensus science information can contribute to social media users' objective and subjective knowledge whereas exposure to anti-consensus information only relates to subjective knowledge.

To test hypotheses 3 to 5 and research question 1, I calculated a structural equation model with maximum likelihood estimation with robust standard errors and a Satorra-Bentler scaled test statistic as some measures were not normally distributed. General science skepticism was modelled as a latent variable with two higher order factors. The frequency of incidental exposure to pro- and anti-science information and knowledge overconfidence were included as manifest variables aggregating all seven science areas. Moreover, because of the large correlation between exposure to pro- and anti-consensus information (see Figure 1), the error terms of both variables were allowed to covary, and I tested for multicollinearity with a linear regression model. The variance inflation factor did not indicate serious problems of multicollinearity (VIF = 4.6). All variables in the model were controlled for participants' age, gender, and education. The fit indices of the SEM indicate a close fit between the model and the observed data: $\chi^2(185) = 528.45$, $p < .001$, $\chi^2/df = 2.86$, CFI = .98, TLI = .97, RMSEA = .038 (90% CI: .035, .041), SRMR = .03. The whole model is depicted in Figure 4.

Hypothesis *H3* assumed a positive relationship between exposure to anti-consensus information and knowledge overconfidence. Indeed, the results revealed such a relationship ($\beta = .30$, $p < .001$) supporting the hypothesis. *RQ1* asked whether incidental exposure to pro-consensus information on social media would be associated with knowledge overconfidence. However, there was no indication of this relationship ($\beta = .02$, $p = .641$). In *H4*, I hypothesized that general science skepticism would be a) negatively associated

with incidental exposure to pro-consensus science information and b) positively to anti-consensus information. The results did not support *H4a*, revealing a relationship that is opposite of the assumed one ($\beta = .24$, $p < .001$). In contrast, *H4b* was supported as the relationship was in the hypothesized direction ($\beta = .26$, $p < .001$). Moreover, *H5* was also supported by the data showing that general science skepticism positively relates to knowledge overconfidence ($\beta = .13$, $p < .001$).

Exploratory Analysis

After being familiar with the results, I was additionally interested in two things which were not part of the preregistration: first, because of the large correlation between incidental exposure to pro- and anti-consensus information, the question emerged how the relationships between both variables and objective science knowledge would change if they were mutually controlled. Second, whether general science skepticism would moderate the relations between pro- and anti-consensus information and objective knowledge. The latter was assumed because persons who are skeptical about science might less likely believe accurate, pro-consensus science information and they might be more susceptible to believe inaccurate, anti-consensus information compared to persons who are less skeptical towards science. For this reason, I calculated a multiple regression analysis with objective knowledge as dependent variable and interaction-terms between exposure to pro- and anti-consensus information and science skepticism. The overall model was significant: $F(5, 1273) = 61.53$, $p < .001$. The data revealed that when controlled for anti-consensus information, there was a medium to large positive relationship between incidental exposure to pro-consensus information and objective knowledge ($\beta = .45$, $p < .001$). Also, when controlled for pro-consensus information, there was a medium negative relation between exposure to anti-consensus information and objective knowledge ($\beta = -.34$, $p < .001$). The interactions with science skepticism, however, were not significant. This indicates that all persons are similarly likely to believe accurate science claims and science misinformation no matter whether they are skeptical towards science or not. The interaction effects can be

seen in Figure 5.

Discussion

The current work had the objectives of studying whether persons with a general negative disposition towards science would be incidentally exposed to more misleading than accurate science content on social media, and whether this would be associated with an overestimation of one's objective knowledge about various science topics. The findings contribute to a better understanding of how general dispositions are associated with incidental exposure to science information and whether people can learn from incidental exposure to accurate scientific information or whether they are misinformed by misleading content.

Science Skepticism and Incidental Exposure to Science Content

It was hypothesized that users who are generally skeptical towards science (i.e., distrust and doubt science and scientists; Meier and Krämer, 2022) might be incidentally exposed to more information that corresponds to their disposition (i.e., information which contradicts scientific consensuses). However, the results revealed that social media users who are skeptical about science stated to be incidentally exposed to both more pro-consensus and more anti-consensus information more frequently. This indicates that science skeptics are not embedded in homogenous information networks. Based on the data of the present study, it is not possible to tell which channels contribute to this composition of accurate science information and misinformation. One possibility to make sense of these findings is that science skeptics self-select into information networks in which science misinformation are prevalent (e.g., by subscribing to certain channels). Based on these behaviors and further interactions, users might be classified as interested in certain science topics (see Thorson et al., 2021) which might algorithmically catalyze incidental exposure to similar content (Barnidge, 2023; Rousseau, 2023). However, it might also be that these algorithmic recommendations happen on a topic- rather than a position-level (e.g., being exposed to content about global warming irrespective of whether the scientific consensus is

correctly outlined or denied) which can explain the positive relation between science skepticism and exposure to both pro- and anti-consensus content. Future works should investigate the sources of incidental exposure to pro- and anti-consensus content as well as those contents particularly disseminated by social media algorithms.

If persons with a skeptical disposition towards science are indeed exposed to both information which is in accordance with and which contradicts scientific consensuses their negative disposition towards science and their false beliefs might be reinforced. Previous studies indicate that science misinformation can easily overwrite positive effects of exposure to accurate science content (Nyhan et al., 2022) and potentially reinforce one's negative attitudes and misdirected views (Southwell et al., 2022; Treen et al., 2020). Future works might take a closer look at the potential effects of a frequent exposure to misleading and false scientific information.

An alternative explanation to make sense of the findings is of methodological nature. Because participants were asked to estimate how frequently they were exposed to pro- and anti-consensus science content in the last three months, their assessments might have been biased based on certain attitudes. Although Vraga and Bode (2020) found positive relations between attention and subsequent recall to different social media contents, they also found that individual interests were positively associated with recall. This means that people are not only recalling attention but also inferring their attitudes. Therefore, future studies should apply different methods to assess incidental exposure like linkage analyses (Otto et al., 2024).

Incidental Exposure to Science Content and Knowledge

Previous studies found that incidental exposure to news or political content can contribute to false perceptions of one's objective knowledge due to increased contact to certain topics without actually receiving novel information (Schäfer, 2020; Yamamoto & Yang, 2022). The results of the present study indicate that social media users who are incidentally exposed to pro-consensus science information from various topics have an

increased subjective knowledge, but they also have a slightly higher objective knowledge (note, however, that there are slight differences on the domain level). These findings expand those by Anderson et al. (2021), who did not find that incidental exposure to information about gene editing relates to more objective knowledge about gene editing. Here, I show that incidental exposure to accurate science content is positively associated with what people know about different science topics which is in accordance with findings from a meta-analysis about incidental exposure to political content (Nanz & Matthes, 2022). Contrarily, users who are incidentally exposed to anti-consensus content were found to have increased subjective knowledge too, but on average there were no relations to their objective knowledge which speaks against the assumption that users who are exposed to false scientific claims are per se misinformed. However, the results also revealed that incidental exposure to pro-consensus content is highly positively associated with incidental exposure to anti-consensus content which indicates that the correlations are confounded. Controlling the relationships between pro-consensus information exposure and objective knowledge for anti-consensus information exposure and vice versa reveals much stronger positive and negative relationships, respectively. This implies that if social media users were only exposed to accurate science information, there was a strong positive relationship with their level of objective knowledge. If, on the other hand, users were only confronted with science misinformation, objective knowledge would be substantially lower. These results are consistent with previous studies which found that exposure to anti-consensus science content is associated with false beliefs (Lee et al., 2020) and that although exposure to accurate science content can contribute to users' accurate knowledge about scientific claims, false information can reverse these effects (Nyhan et al., 2022). As the present study found that users who are more frequently exposed to accurate science information also seem to be more frequently exposed to false science information, they might feel uncertain about what is true and what is false and gains in objective knowledge might be reduced or even absent. Future works could build up on these results and investigate user attributes or

information sources which contribute to or impair user beliefs about scientific consensuses.

Finally, the present study found no support for the claim that incidental exposure to pro-consensus information relates to knowledge overconfidence which has been found regarding news or political content (Schäfer, 2020; Yamamoto & Yang, 2022)). Hence, it seems that users who are frequently exposed to accurate science information not only have higher objective knowledge but can also better subjectively assess their knowledge levels. Contrarily, persons who see more anti-consensus information on social media tend to overestimate their knowledge probably because they are misinformed but believe that they receive accurate information (see Anspach et al., 2019). Since overestimating one's science knowledge has been found to be associated with an increased likelihood to share misinformation and a reduced public support for science (Ho et al., 2019; Motta et al., 2018), the findings of the present study have important implications for handling false scientific information on social media. Social media companies should increase their efforts to minimize the chances of being exposed to scientific misinformation as people potentially remember wrong and misleading scientific information (Greene et al., 2021) and become uncertain about whether accurate science information are true (Nyhan et al., 2022).

Limitations and Future Directions

When interpreting the results of the present study, it is important to consider its limitations. First, the data are of cross-sectional nature which means that they do not allow to draw causal conclusions. Future works should assess incidental exposure to science content over multiple measurement waves to be able to observe longitudinal dynamics. Second, the measure of incidental exposure is based on self-reports which are likely prone to attention or memory biases (Vraga & Bode, 2020). Although self-reports are a common method to examine incidental exposure (Nanz & Matthes, 2022), the findings of the present study should be understood as initial evidence of incidental exposure to science content, and I strongly encourage other researchers to rely on more sophisticated and objective techniques in future studies (e.g., by means of data tracking or donations; Otto

et al., 2024). Third, the results might be limited due to the German or European context and strict content regulation such as the GDPR. Therefore, in other countries, social media companies might feed algorithms with even more user data than in the EU. Fourth, I used three items per science topic to assess participants' objective knowledge of each of the seven science topics. Evidently, this only constitutes an excerpt of and an approximation to the knowledge that can be acquired in each of the domains. Fifth, while this work has shown that a general (negative) disposition towards science can be positively related to incidental exposure with accurate and false science information, it does not reveal anything about user engagement with such information. Hence, future works might pursue the questions if and how users interact (e.g., read, like, or share) with pro- or anti-consensus science information.

Conclusion

On social media sites, users are incidentally exposed to contents that are shared by social contacts, disseminated by subscribed channels, or recommended by algorithms. The present work pursued the question whether social media users with a skeptical disposition towards science would be exposed to more anti- than pro-consensus information and whether that would be associated with a tendency to overestimate one's objective science knowledge. The findings showed that persons who are skeptical about science are more likely to be exposed to both pro- as well as anti-consensus information. This indicates that these persons are not necessarily embedded in homogenous networks and that they might have been algorithmically classified as generally interested in science topics, irrespective of their negative disposition towards science. Users who are more frequently exposed to pro-consensus information were found to have increased objective knowledge which indicates that even non-intentional exposure to accurate science information can be beneficial for learning about science. Contrarily, those who are primarily exposed to anti-consensus science information tended to overestimate their objective knowledge which indicates that the users are misinformed. Altogether, these results show that general

dispositions towards science can be associated with incidental exposure to science content and that even unintentional contact with science information on social media might positively contribute to people's basic knowledge about science topics. Nonetheless, as more frequent incidental exposure to anti-consensus content is associated with lowered objective knowledge as shown in this study, it is important that social media companies and regulators continue their efforts in reducing misleading and false content on social media.

References

Ahmadi, M., & Wohn, D. Y. (2018). The antecedents of incidental news exposure on social media. *Social Media + Society*, *4*(2), 205630511877282. https://doi.org/10.1177/2056305118772827

Allgaier, J. (2019). Science and environmental communication on YouTube: Strategically distorted communications in online videos on climate change and climate engineering. *Frontiers in Communication*, *4*, 36. https://doi.org/10.3389/fcomm.2019.00036

Anderson, J. T. L., Howell, E. L., Xenos, M. A., Scheufele, D. A., & Brossard, D. (2021). Learning without seeking?: Incidental exposure to science news on social media & knowledge of gene editing. *Journal of Science Communication*, *20*(4), A01. https://doi.org/10.22323/2.20040201

Anspach, N. M., Jennings, J. T., & Arceneaux, K. (2019). A little bit of knowledge: Facebook's news feed and self-perceptions of knowledge. *Research & Politics*, *6*(1), 205316801881618. https://doi.org/10.1177/2053168018816189

Barnidge, M. (2023). Incidental exposure and news engagement: Testing temporal order and the role of political interest. *Digital Journalism*, *11*(1), 125–143. https://doi.org/10.1080/21670811.2021.1906290

Betsch, C., & Wicker, S. (2012). E-health use, vaccination knowledge and perception of own risk: Drivers of vaccination uptake in medical students. *Vaccine*, *30*(6), 1143–1148. https://doi.org/10.1016/j.vaccine.2011.12.021

Brucks, M. (1985). The effects of product class knowledge on information search behavior. *Journal of Consumer Research*, *12*(1), 1. https://doi.org/10.1086/209031

Chen, H.-T., Kim, Y., & Chan, M. (2022). Just a glance, or more? pathways from counter-attitudinal incidental exposure to attitude (de)polarization through response behaviors and cognitive elaboration. *Journal of Communication*, *72*(1), 83–110. https://doi.org/10.1093/joc/jqab046

578 Cologna, V., Mede, N., Berger, S., Besley, J., Brick, C., Joubert, M., Maibach, E.,
579 Mihelj, S., Oreskes, N., Schäfer, M., et al. (2024). Trust in scientists and their role
580 in society across 67 countries. https://doi.org/10.31234/osf.io/x9jng
581 DeVito, M. A. (2017). From editors to algorithms: A values-based approach to
582 understanding story selection in the facebook news feed. *Digital Journalism*, *5*(6),
583 753–773. https://doi.org/10.1080/21670811.2016.1178592
584 Greene, J. A., Copeland, D. Z., & Deekens, V. M. (2021). A model of technology incidental
585 learning effects. *Educational Psychology Review*, *33*(3), 883–913.
586 https://doi.org/10.1007/s10648-020-09575-5
587 Ho, S. S., Looi, J., Leong, A. D., & Leung, Y. W. (2019). Explicating factual and
588 subjective science knowledge: Knowledge as a mediator of news attention and
589 attitudes. *Asian Journal of Communication*, *29*(1), 73–91.
590 https://doi.org/10.1080/01292986.2018.1518466
591 Huber, B., Barnidge, M., Gil De Zúñiga, H., & Liu, J. (2019). Fostering public trust in
592 science: The role of social media. *Public Understanding of Science*, *28*(7), 759–777.
593 https://doi.org/10.1177/0963662519869097
594 Kruger, J., & Dunning, D. (1999). Unskilled and unaware of it: How difficulties in
595 recognizing one's own incompetence lead to inflated self-assessments. *Journal of*
596 *Personality and Social Psychology*, *77*(6), 1121–1134.
597 https://doi.org/10.1037/0022-3514.77.6.1121
598 Lakens, D. (2022). Sample size justification. *Collabra: Psychology*, *8*(1), 33267.
599 https://doi.org/10.1525/collabra.33267
600 Lee, J. J., Kang, K.-A., Wang, M. P., Zhao, S. Z., Wong, J. Y. H., O'Connor, S.,
601 Yang, S. C., & Shin, S. (2020). Associations between COVID-19 misinformation
602 exposure and belief with COVID-19 knowledge and preventive behaviors:
603 Cross-sectional online study. *Journal of Medical Internet Research*, *22*(11), e22205.
604 https://doi.org/10.2196/22205

Lewandowsky, S., Gignac, G. E., & Oberauer, K. (2013). The role of conspiracist ideation and worldviews in predicting rejection of science (T. Denson, Ed.). *PLoS ONE*, *8*(10), e75637. https://doi.org/10.1371/journal.pone.0075637

Light, N., Fernbach, P. M., Rabb, N., Geana, M. V., & Sloman, S. A. (2022). Knowledge overconfidence is associated with anti-consensus views on controversial scientific issues. *Science Advances*, *8*(29), eabo0038. https://doi.org/10.1126/sciadv.abo0038

Matthes, J., Nanz, A., Stubenvoll, M., & Heiss, R. (2020). Processing news on social media. the political incidental news exposure model (PINE). *Journalism*, *21*(8), 1031–1048. https://doi.org/10.1177/1464884920915371

Meier, Y., & Krämer, N. (2022, July 15). General science skepticism: Development and validation of a new scale. https://doi.org/10.31234/osf.io/x9jng

Motta, M., Callaghan, T., & Sylvester, S. (2018). Knowing less but presuming more: Dunning-kruger effects and the endorsement of anti-vaccine policy attitudes. *Social Science & Medicine*, *211*, 274–281. https://doi.org/10.1016/j.socscimed.2018.06.032

Nanz, A., & Matthes, J. (2022). Democratic consequences of incidental exposure to political information: A meta-analysis. *Journal of Communication*, *72*(3), 345–373. https://doi.org/10.1093/joc/jqac008

Nyhan, B., Porter, E., & Wood, T. J. (2022). Time and skeptical opinion content erode the effects of science coverage on climate beliefs and attitudes. *Proceedings of the National Academy of Sciences*, *119*(26), e2122069119. https://doi.org/10.1073/pnas.2122069119

Oeldorf-Hirsch, A. (2018). The role of engagement in learning from active and incidental news exposure on social media. *Mass Communication and Society*, *21*(2), 225–247. https://doi.org/10.1080/15205436.2017.1384022

Otto, L. P., Loecherbach, F., & Vliegenthart, R. (2024). Linkage analysis revised–linking digital traces and survey data. *Communication Methods and Measures*, *18*(2), 186–204.

Park, C. W., & Lessig, V. P. (1981). Familiarity and its impact on consumer decision biases and heuristics. *Journal of Consumer Research*, *8*(2), 223. https://doi.org/10.1086/208859

Partheymüller, J., Kritzinger, S., & Plescia, C. (2022). Misinformedness about the european union and the preference to vote to leave or remain. *JCMS: Journal of Common Market Studies*, *60*(5), 1449–1469. https://doi.org/10.1111/jcms.13316

Pavelle, S., & Wilkinson, C. (2020). Into the digital wild: Utilizing twitter, instagram, YouTube, and facebook for effective science and environmental communication. *Frontiers in Communication*, *5*, 575122. https://doi.org/10.3389/fcomm.2020.575122

Plohl, N., & Musil, B. (2018). Do i know as much as i think i do? the dunning-kruger effect, overclaiming, and the illusion of knowledge. *Psihološka obzorja / Horizons of Psychology*, 20–30. https://doi.org/10.20419/2018.27.481

Rousseau, A. (2023). Reciprocal relationships between adolescents' incidental exposure to climate-related social media content and online climate change engagement. *Communication Research*, 009365022311646. https://doi.org/10.1177/00936502231164675

Sabir, A., Lafontaine, E., & Das, A. (2022). Analyzing the impact and accuracy of facebook activity on facebook's ad-interest inference process. *Proceedings of the ACM on Human-Computer Interaction*, *6*, 1–34. https://doi.org/10.1145/3512923

Schäfer, S. (2020). Illusion of knowledge through facebook news? effects of snack news in a news feed on perceived knowledge, attitude strength, and willingness for discussions. *Computers in Human Behavior*, *103*, 1–12. https://doi.org/10.1016/j.chb.2019.08.031

Southwell, B. G., Brennen, J. S. B., Paquin, R., Boudewyns, V., & Zeng, J. (2022). Defining and measuring scientific misinformation. *The ANNALS of the American Academy of Political and Social Science*, *700*(1), 98–111. https://doi.org/10.1177/00027162221084709

659 Taddicken, M., & Krämer, N. (2021). Public online engagement with science information:
660 On the road to a theoretical framework and a future research agenda. *Journal of*
661 *Science Communication*, *20*(3), A05. https://doi.org/10.22323/2.20030205
662 Takahashi, B., & Tandoc, E. C. (2016). Media sources, credibility, and perceptions of
663 science: Learning about how people learn about science. *Public Understanding of*
664 *Science*, *25*(6), 674–690. https://doi.org/10.1177/0963662515574986
665 Tewksbury, D., Weaver, A. J., & Maddex, B. D. (2001). Accidentally informed: Incidental
666 news exposure on the world wide web. *Journalism & Mass Communication*
667 *Quarterly*, *78*(3), 533–554. https://doi.org/10.1177/107769900107800309
668 Thorson, K. (2020). Attracting the news: Algorithms, platforms, and reframing incidental
669 exposure. *Journalism*, *21*(8), 1067–1082. https://doi.org/10.1177/1464884920915352
670 Thorson, K., Cotter, K., Medeiros, M., & Pak, C. (2021). Algorithmic inference, political
671 interest, and exposure to news and politics on facebook. *Information,*
672 *Communication & Society*, *24*(2), 183–200.
673 https://doi.org/10.1080/1369118X.2019.1642934
674 Thorson, K., & Wells, C. (2016). Curated flows: A framework for mapping media exposure
675 in the digital age: Curated flows. *Communication Theory*, *26*(3), 309–328.
676 https://doi.org/10.1111/comt.12087
677 Tian, Y., & Robinson, J. D. (2009). Incidental health information use on the internet.
678 *Health Communication*, *24*(1), 41–49. https://doi.org/10.1080/10410230802606984
679 Tobler, C., Visschers, V. H. M., & Siegrist, M. (2012). Consumers' knowledge about
680 climate change. *Climatic Change*, *114*(2), 189–209.
681 https://doi.org/10.1007/s10584-011-0393-1
682 Treen, K. M. d., Williams, H. T. P., & O'Neill, S. J. (2020). Online misinformation about
683 climate change. *WIREs Climate Change*, *11*(5). https://doi.org/10.1002/wcc.665

Vraga, E. K., & Bode, L. (2020). Defining misinformation and understanding its bounded nature: Using expertise and evidence for describing misinformation. *Political Communication*, *37*(1), 136–144. https://doi.org/10.1080/10584609.2020.1716500

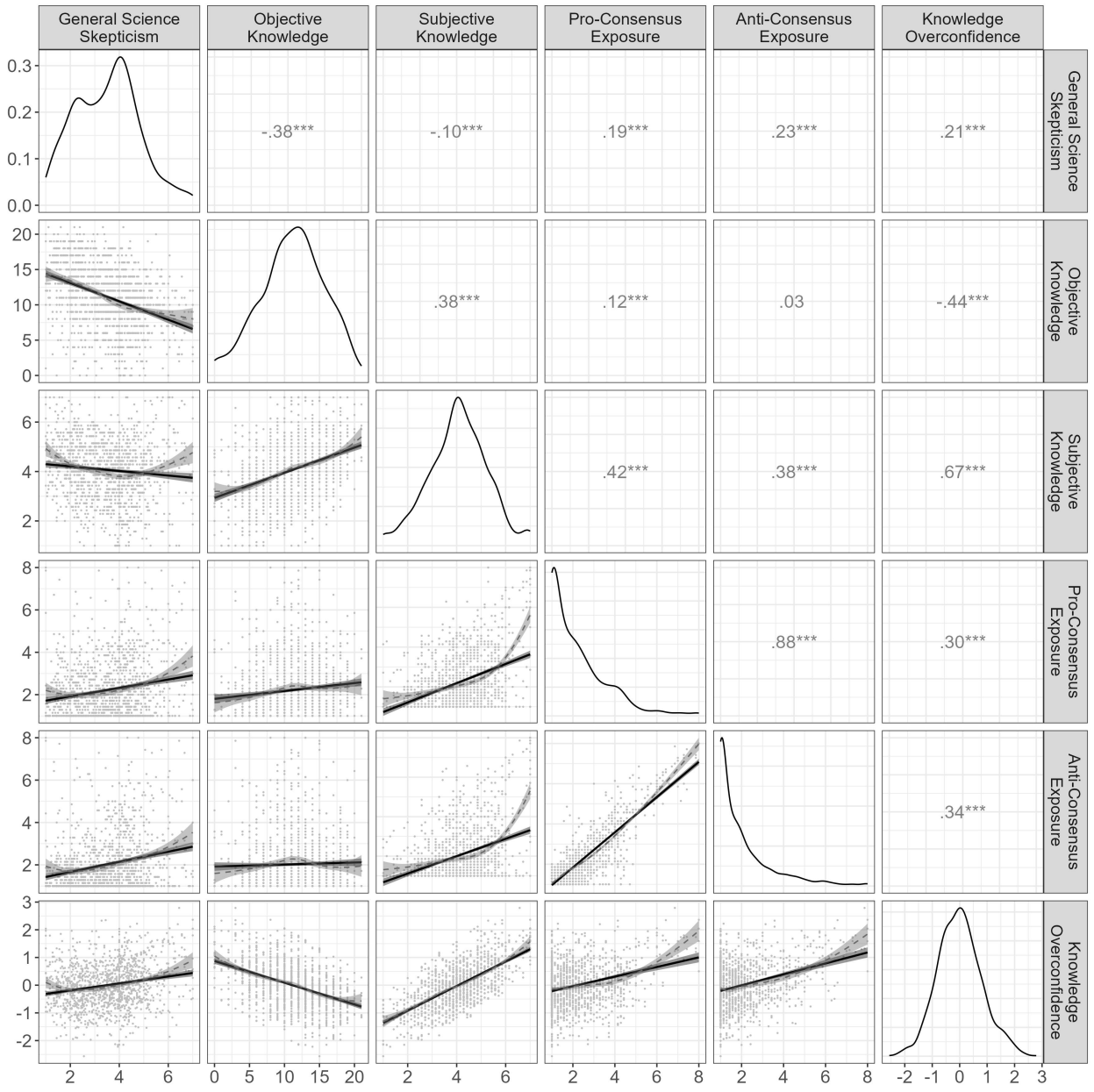
Walter, S., Brüggemann, M., & Engesser, S. (2018). Echo chambers of denial: Explaining user comments on climate change. *Environmental Communication*, *12*(2), 204–217. https://doi.org/10.1080/17524032.2017.1394893

Williams Kirkpatrick, A. (2021). The spread of fake science: Lexical concreteness, proximity, misinformation sharing, and the moderating role of subjective knowledge. *Public Understanding of Science*, *30*(1), 55–74. https://doi.org/10.1177/0963662520966165

Yamamoto, M., & Yang, F. (2022). Does news help us become knowledgeable or think we are knowledgeable? examining a linkage of traditional and social media use with political knowledge. *Journal of Information Technology & Politics*, *19*(3), 269–283. https://doi.org/10.1080/19331681.2021.1969611

Figure 1

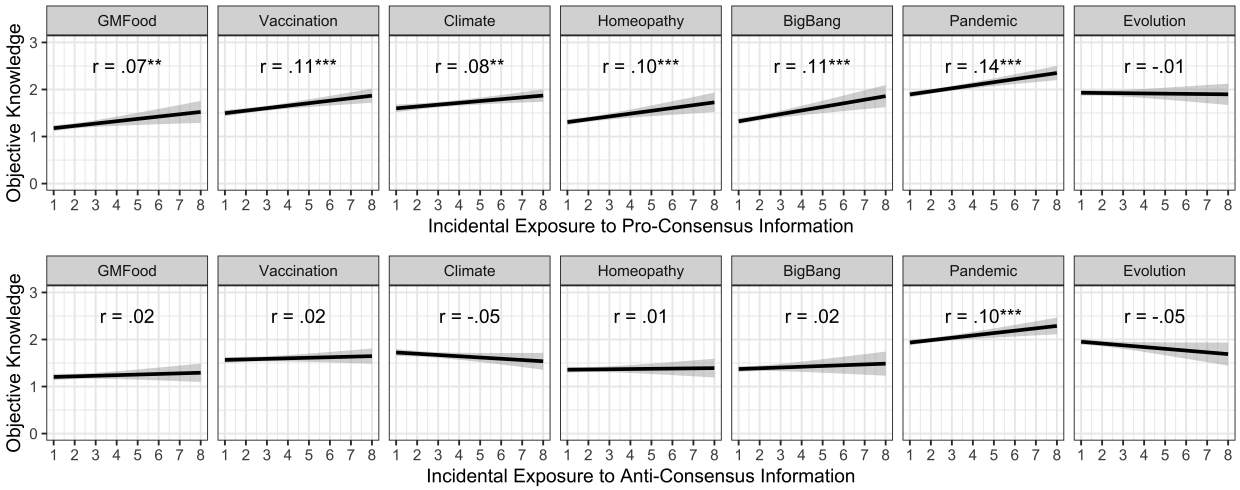
Zero-Order Correlations of the Main Variables



Note. Above diagonal: zero-order correlations (***$p < .001$). Diagonal: density plots. Below diagonal: scatterplots of the zero-order correlations with regression lines (solid lines) and quadratic regression lines (dotted lines).

Figure 2

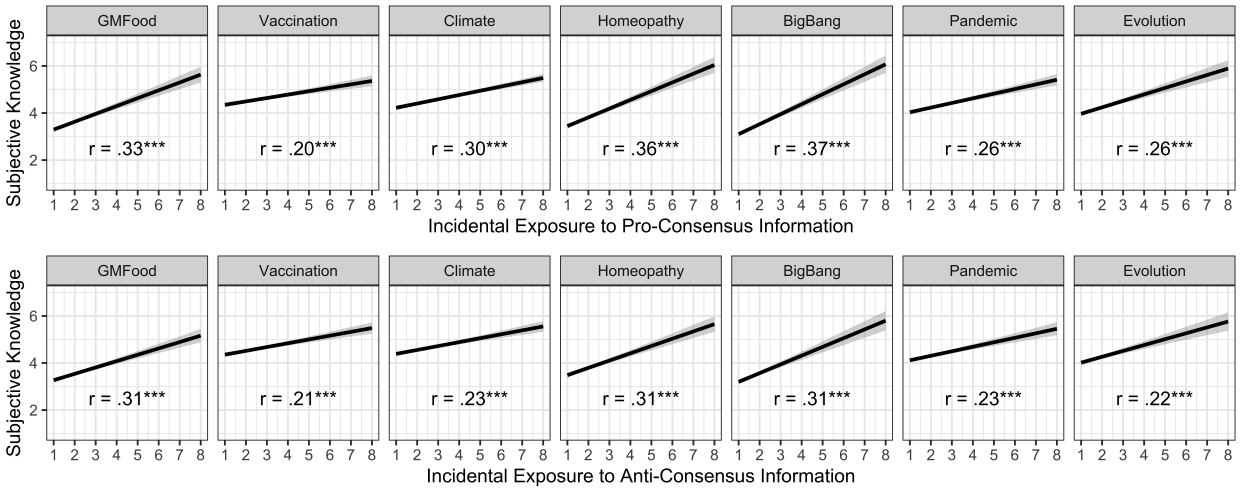
Relationships between Incidental Exposure to Pro- or Anti-Consensus Information and Objective Knowledge



Note. $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$. SESOI = |.10|

Figure 3

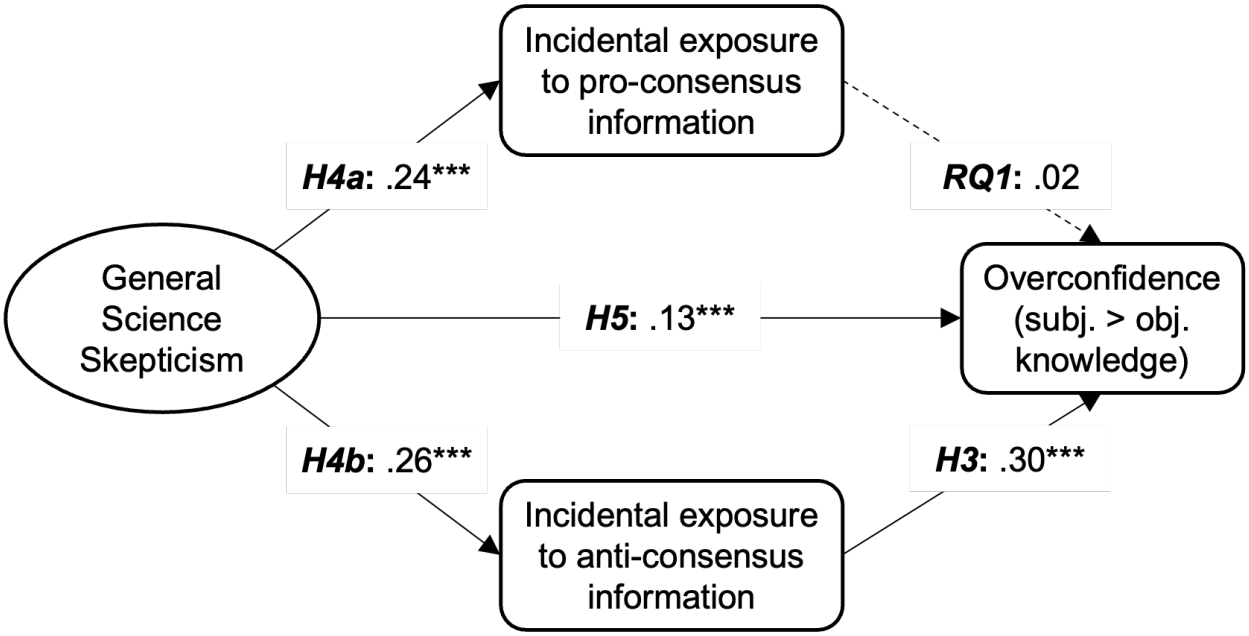
Relationships between Incidental Exposure to Pro- or Anti-Consensus Information and Subjective Knowledge



Note. $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$. SESOI = |.10|

Figure 4

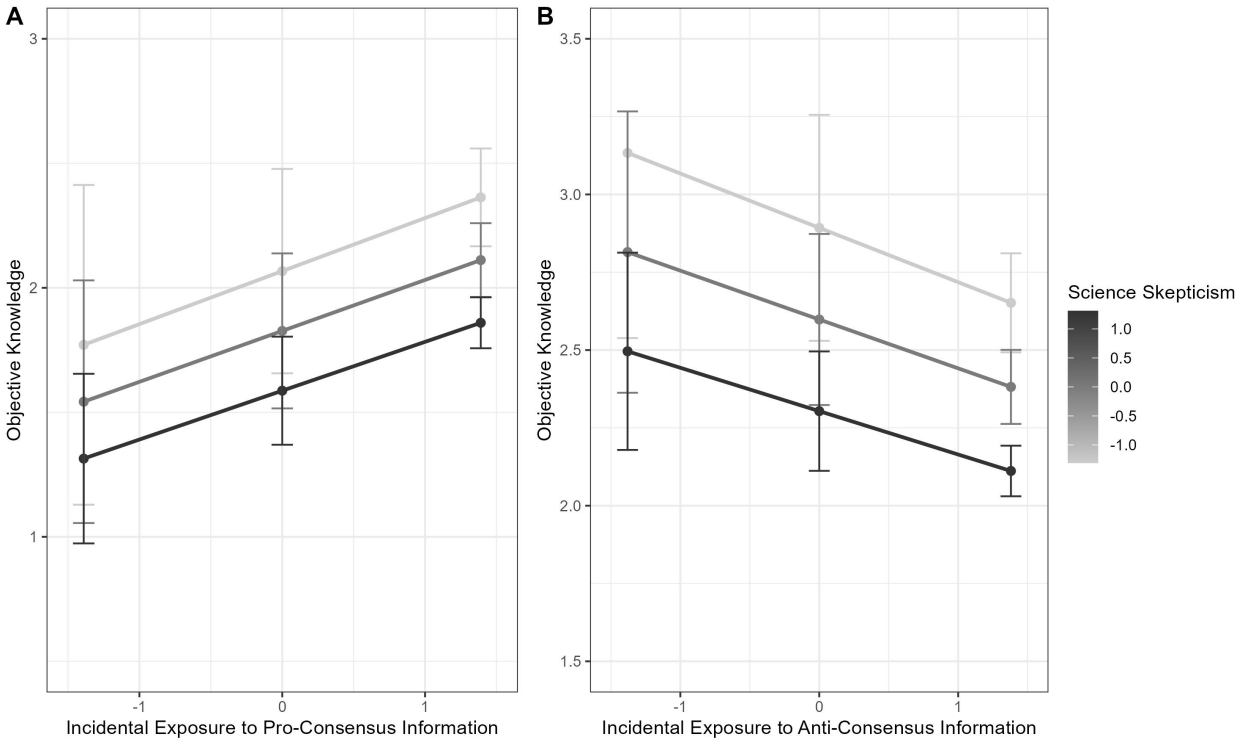
Structural Equation Model controlled for Age, Gender, and Education



Note. Numbers are the standardized regression coefficients. Variables in circles were modelled latently, variables in squares were modelled manifestly. Model-fit: $\chi^2(185) = 528.45$, $p < .001$, $\chi^2/df = 2.86$, CFI = .98, TLI = .97, RMSEA = .038 (90% CI: .035, .041), SRMR = .03. $^{***}p < .001$

Figure 5

Interaction Plots



Note. Moderating effect of general science skepticism on the relations between (A) incidental exposure to pro-consensus information and objective science knowledge and (B) incidental exposure to anti-consensus information and objective science knowledge. The interaction effects are not significant.